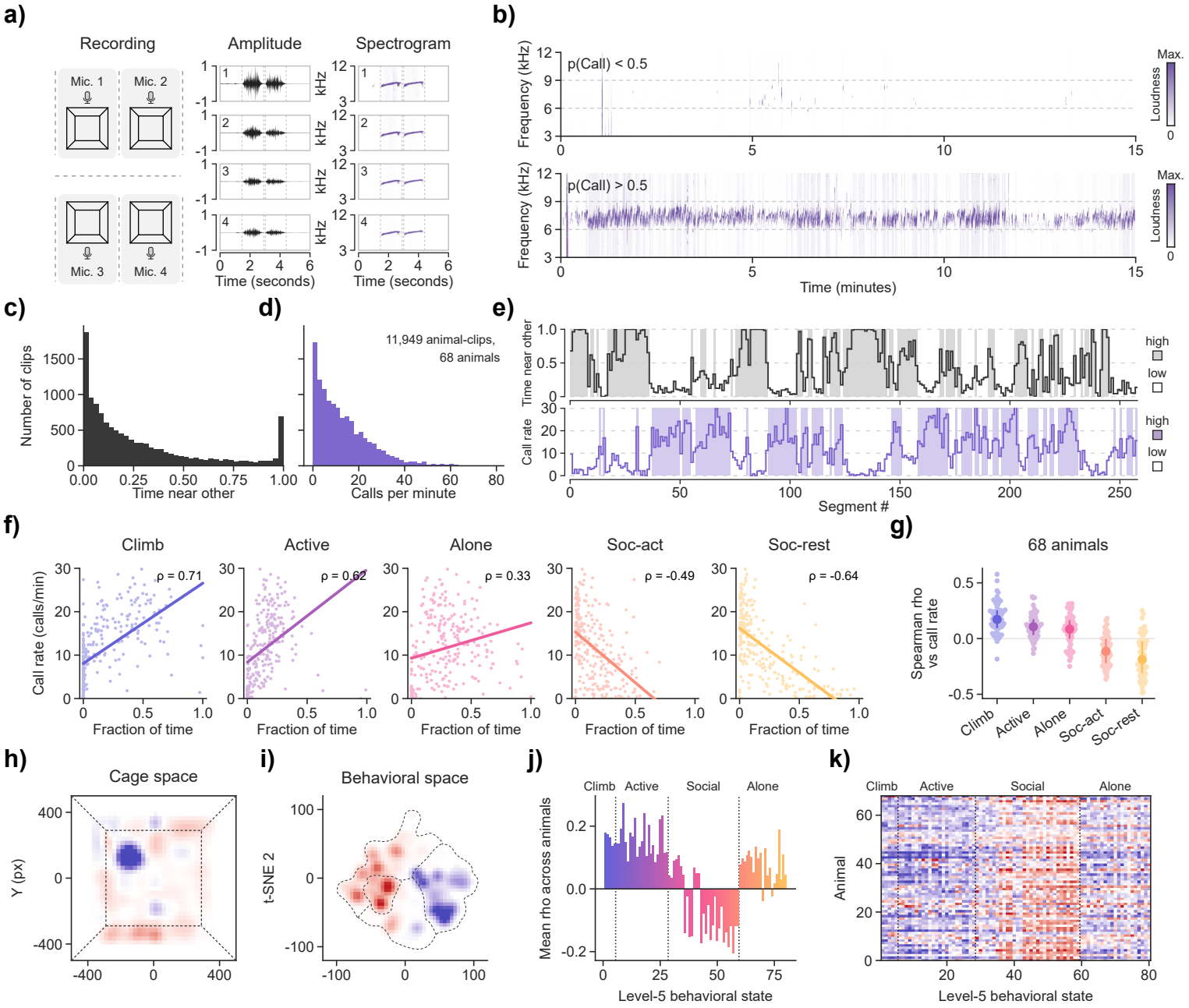

a)
Recording
Amplitude
Spectrogram
Mic. 1
Mic. 2
Mic. 3
Mic. 4
kHz
Time (seconds)
b)
p(Call) < 0.5
p(Call) > 0.5
Frequency (kHz)
Loudness
Max.
Time (minutes)
c)
Number of clips
Time near other
d)
11,949 animal-clips,
68 animals
Calls per minute
e)
Time near other
Call rate
high
low
Segment #
f)
Climb
Active
Alone
Soc-act
Soc-rest
Call rate (calls/min)
ρ = 0.71
ρ = 0.62
ρ = 0.33
ρ = -0.49
ρ = -0.64
Fraction of time
g)
68 animals
Spearman rho
vs call rate
h)
Cage space
Y (px)
i)
Behavioral space
t-SNE 2
j)
Climb
Active
Social
Alone
Mean rho across animals
Level-5 behavioral state
k)
Animal
Level-5 behavioral state